SHUBHAM MODH

.NET Developer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Full-Stack .NET Developer with **2.9+ years of production experience** building real-time transit and fleet-tracking platforms for state and municipal transport corporations. Specialized in **ASP.NET Core Web API** backends, **PostgreSQL** (stored procedures, partitioning, query optimization), and **Angular** front-ends, with event-driven pipelines (**Kafka, Redis, Tile38**) and GPS/OBU device-protocol integration. Primary developer on **AVLS (Automatic Vehicle Location System)** — live vehicle tracking for **~700 buses** across NMMT, ASTC, and OSRTC deployments.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL (PostgreSQL)
Backend: ASP.NET Core (.NET Core 3.1), Web API, microservices & event-driven architecture, Entity Framework Core, ADO.NET, LINQ, TCP socket programming, SignalR
Front-End: Angular, Kendo UI, Google Maps API
Databases: PostgreSQL — stored procedures, partitioning, execution-plan tuning, Foreign Data Wrappers (FDW), logical replication; MongoDB
Data & Messaging: Apache Kafka, Redis, Tile38 (geospatial)
Infrastructure: Linux, Google Cloud Platform, Docker, Jenkins CI/CD, Kibana, Git (GitHub, GitLab)
Security: Keycloak, IDAM, JWT-based authentication
Domain: GPS/AVL telemetry & device-protocol parsing, geofencing, trip & ETA computation, real-time fleet tracking, Passenger Information Systems (PIS), public-transport ITMS

PROFESSIONAL EXPERIENCE

Associate Software Developer — Amnex Infotechnologies, Ahmedabad *Feb 2024 – Present*

- Owned and delivered the **AVLS (Automatic Vehicle Location System)** module across **municipal and state transport deployments (NMMT, ASTC)** — ASP.NET Core APIs, Angular front-end, PostgreSQL database, and production support.
- Built **GPS / OBU device panels** parsing vendor protocol frames over **raw TCP** across multiple vendors and firmware variants, feeding **Kafka, PostgreSQL, Redis, and Tile38** for real-time tracking and geofencing.
- Built and extended the **Masters module** across **OSRTC, ASTC, and NMMT** — route & geofence creation via the **Google Maps API**, depot / vehicle / employee / user management, **Keycloak / IDAM** login, and per-client role-based menus.
- Delivered full-stack features across **OPRS** (Online Passenger Revenue System — **50+ transactional reports**), **GRMS**, and **DMS**; implemented the **PIS (Passenger Information System)** display panel on hardware from two vendors.
- Handled **production support and performance tuning** across the stack — PostgreSQL partitioning and query tuning, Kafka pipeline reliability, and .NET service optimization.

KEY PROJECTS

NMMT — Navi Mumbai Municipal Transport | *ASP.NET Core, Angular, PostgreSQL, Kafka, Redis, Tile38, SignalR*

*Municipal bus operations for Navi Mumbai — **~700 buses live-tracked**, one GPS packet per vehicle every ~10 seconds. Modules: AVLS, PIS, Masters, Reports.*

- Own the **AVLS module end-to-end** — the **live tracking map** (vehicle positions, status, movement trails), **trip management** (scheduled vs actual trips, trip start/close, schedule adherence), and **geofence-based stop-arrival detection**.
- Developed **ETA computation and display** for upcoming stops, **route replay** of completed trips, and **real-time incident detection** (overspeed, route violation, bus bunching, missed stop, panic) with a control-room alert panel.
- Built the **emergency Panic Alert system** (database trigger + **SignalR real-time push** + global on-screen alert) and a **Redis-backed enrichment layer** for fast live tracking.
- Maintain the module in production — **partitioned high-volume telemetry/incident history tables**, **tuned stored procedures** behind the live dashboards, and **logical replication** for master-data tables.

ASTC (Assam) & OSRTC (Odisha) — State Transport Deployments | *ASP.NET Core, Angular, PostgreSQL, Kafka*

- Delivered the **AVLS tracking stack** for ASTC — including the **Emergency Panic Alert** feature later ported to NMMT — and integrated **PostgreSQL Foreign Data Wrappers (FDW)** for cross-database master data.
- Built the **Masters module** for OSRTC — route & geofence creation on Google Maps API and depot, vehicle, employee, and user management with role-based menu assignment.

EDUCATION

B.E., Computer Engineering — Vishwakarma Government Engineering College, Ahmedabad · CGPA 8.12/10 *2021 – 2024*
Diploma, Computer Engineering — Ganpat University, Mehsana · CGPA 9.48/10 *2018 – 2021*